`microimpute`: Benchmarking Cross-Survey Imputation Methods for U.S. Household Wealth

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Abstract

Microdata surveys often lack variables critical for policy analysis, requiring imputation from richer donor surveys, yet researchers have lacked standardized tools for systematically comparing imputation methods and selecting the best approach for a given dataset. We introduce `microimpute`, an open-source Python package that provides a unified framework for benchmarking, tuning, and automatically selecting among multiple imputation methods. We apply it to impute U.S. household wealth from the Survey of Consumer Finances (SCF) onto the Current Population Survey (CPS). Evaluating five methods—Quantile Random Forests (QRF), Ordinary Least Square (OLS), Quantile Regression, Hot Deck Matching, and Mixture Density Networks, we find that QRF achieves the lowest quantile loss and Wasserstein distance, reflecting its capacity to model the non-linear relationships between demographic predictors and wealth. A downstream Supplemental Security Income simulation demonstrates the practical stakes of method choice: without imputed wealth, the microsimulation overestimates SSI recipients by 167%, while all five imputation methods substantially reduce this gap. Cross-dataset benchmarking across six additional domains reveals, however, that no single method dominates universally. Although QRF proves best at capturing complex, non-linear relationships, Hot Deck Matching better preserves marginal distributions by drawing directly from the donor pool, while OLS remains competitive for approximately linear relationships. These findings underscore that imputation method selection should be empirically driven rather than prescribed, motivating standardized benchmarking tools, like `microimpute`, that enable researchers to identify the best approach for their specific data characteristics and research objectives.

1 Introduction

Statistical matching, also known as data fusion or full variable imputation, addresses the challenge of combining information from multiple data sources that share common variables but contain different samples of units (D'Orazio et al., 2006), which is commonly found

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across fields in empirical research. In its simplest form, the problem involves a donor dataset containing both shared and unique variables and a receiver dataset containing only the shared variables, where the goal is to impute the missing variables for observations in the receiver file based on their shared characteristics (D'Orazio et al., 2021). This framework generalizes traditional missing data imputation by recognizing that "missingness" can arise not only from nonresponse within a single survey but also from the structural absence of variables across distinct data sources.

The relevance of statistical matching extends across the social sciences, particularly in microsimulation modeling where policy analysis requires combining detailed demographic information from one survey with economic variables from another (Bourguignon and Spadaro, 2006). For instance, household surveys may capture income and consumption patterns but lack wealth data, while financial surveys provide wealth information for different samples. Constructing synthetic files that combine these variables enables richer policy analysis than either source alone permits.

The central methodological challenge is accurately modeling the conditional distribution of the target variable given shared predictors, as different approaches make different assumptions about this relationship. Parametric methods may be too restrictive for economic variables exhibiting heavy tails and heteroscedasticity, while nonparametric and machine learning approaches offer greater flexibility at the cost of additional complexity (Meinshausen, 2006; Andridge and Little, 2010). Section 2 reviews these trade-offs in detail.

This paper presents the `microimpute` package, a Python library implementing five statistical matching methods: Hot Deck Matching, Ordinary Least Squares, Quantile Regression, Mixture Density Networks, and Quantile Random Forests. The package provides a comprehensive framework for comparing these methods through systematic benchmarking, with automated model selection based on quantile loss performance. We demonstrate the methodology through an application to wealth imputation, transferring net worth data from the Survey of Consumer Finances to the Current Population Survey. This task exemplifies the challenges of statistical matching for heavy-tailed distributions.

This work provides the following contributions:

- An open-source implementation of a unified framework for five imputation methods, facilitating systematic comparison across methods and support statistical matching through accessible, user-friendly software
- An automated benchmarking pipeline that evaluates distributional accuracy using quantile loss, enabling researchers to select the most appropriate method for their specific data characteristics
- Empirical evidence on the relative performance of these methods for wealth imputation, suggesting that Quantile Random Forests are well-suited to heavy-tailed distributions while highlighting the context-dependence of method choice

The remainder of this paper is organized as follows. Section 2 reviews the statistical matching problem, including its formal definition, the conditional independence assumption that underlies all matching methods, and the five imputation approaches implemented in `microimpute`. Section 3 describes our data sources. Section 4 presents the package architecture and benchmarking methodology. Section 5 reports empirical results from the wealth imputation application. Section 6 discusses implications and limitations, and Section 7 concludes.

2 Background

This section establishes the theoretical foundations for statistical matching and reviews the imputation methods implemented in `microimpute`. We begin with the formal problem definition and the key assumption underlying all statistical matching procedures, then discuss the relationship to missing data mechanisms, and finally describe each of the five imputation methods.

2.1 The Statistical Matching Problem

Statistical matching arises when a researcher seeks to combine information from two (or more) data sources that have been collected independently on different samples (D'Orazio et al., 2006). The canonical setup involves:

- A **donor file** $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n_D}$ containing n_D observations with common variables X and target variable(s) Y
- A **receiver file** $\mathcal{R} = \{x_j\}_{j=1}^{n_R}$ containing n_R observations with only the common variables X

The goal is to impute values $\hat{y}_j$ for each observation j in the receiver file, creating a synthetic file that combines the variables from both sources. When the receiver file also contains variables Z not present in the donor file, the resulting synthetic file contains all three sets of variables (X, Y, Z), though Y and Z are never jointly observed.

Formally, statistical matching seeks to estimate the conditional distribution $P(Y|X)$ from the donor data and use it to generate imputations for the receiver observations. Let $\hat{P}(Y|X = x)$ denote an estimate of this conditional distribution. For a receiver observation with covariates x_j, we can either:

1. Predict a specific quantile, such as the conditional median: $\hat{y}_j = \hat{Q}_{0.5}(Y|X = x_j)$
2. Draw a random value from the estimated distribution: $\hat{y}_j \sim \hat{P}(Y|X = x_j)$

The first approach (deterministic imputation) may be appropriate when only point predictions are needed, though it systematically underestimates variance (Rubin, 1987). The second approach (stochastic imputation) preserves the variability of Y and is preferred when distributional properties must be maintained. This is often the case in microsimulation modeling, where full distributional policy analysis is required.

2.2 The Conditional Independence Assumption

All statistical matching procedures rely, either explicitly or implicitly, on the Conditional Independence Assumption (CIA):

$$Y \perp Z \mid X \tag{1}$$

This assumption states that the target variable Y (from the donor) and any variables Z unique to the receiver are independent, conditional on the common variables X (D'Orazio et al., 2006). In other words, once we account for the shared covariates, knowing Z provides no additional information about Y.

The CIA is fundamentally untestable because Y and Z are never jointly observed in either data source. Its plausibility depends on the richness of the common variables X. When X captures the key determinants of both Y and Z, the assumption is more likely to hold. In practice, researchers must rely on substantive knowledge to assess whether the available common variables are sufficient to render Y and Z conditionally independent.

Violations of the CIA lead to biased estimates of the joint distribution of (Y, Z) in the synthetic file. However, if the research question concerns only the marginal distribution of Y in the receiver population, the CIA is not strictly required. What matters is that the conditional distribution $P(Y|X)$ is correctly specified and that X has sufficient overlap between donor and receiver files (D'Orazio et al., 2021).

2.3 Missing Data Mechanisms

Statistical matching may also warrant discussion within the framework of missing data theory, where Y is "missing" for all observations in the receiver file (Little and Rubin, 2002). The missing data literature distinguishes three mechanisms:

- **Missing Completely at Random (MCAR)**: The probability of missingness is unrelated to both observed and unobserved variables. In the statistical matching context, this would require that selection into the donor versus receiver sample is independent of all variables.

- **Missing at Random (MAR)**: The probability of missingness depends only on observed variables. For statistical matching, this means that conditional on X, the probability of being in the donor sample (and thus having Y observed) does not depend on the value of Y itself.

- **Missing Not at Random (MNAR)**: The probability of missingness depends on the unobserved values. This would occur if, for example, high-wealth individuals were systematically less likely to appear in the wealth survey.

Under MAR, valid inference about $P(Y|X)$ can be obtained from the donor data alone, which justifies the statistical matching approach. Most imputation methods, including those implemented in `microimpute`, assume MAR. When MNAR is suspected, sensitivity analyses or methods that explicitly model the selection mechanism may be required (Little and Rubin, 2002).

In statistical matching applications, an important consideration arises when the donor and receiver files come from surveys with different sampling designs. For instance, wealth surveys often oversample high-net-worth households to improve precision for this rare population, while general household surveys use designs representative of the broader population. This differential sampling can introduce a form of selection that resembles MNAR if not properly addressed. Luckily, survey weights provide a mechanism to mitigate this issue. By incorporating weights that reflect each observation's representativeness of the target population, imputation methods can adjust for known differences in sampling probabilities between the donor and receiver surveys (DuGoff et al., 2014). The `microimpute` package supports the use of survey weights throughout its imputation pipeline, allowing methods to account for complex survey designs. While survey weights cannot fully address MNAR when missingness depends on unobserved values of Y itself, they can substantially reduce bias arising from differential sampling designs across data sources.

2.4 Imputation Methods

We now describe the five imputation methods implemented in `microimpute`. Each method provides a different approach to estimating the conditional distribution $P(Y|X)$ and generating imputations.

2.4.1 Hot Deck Matching

Hot deck imputation replaces missing values in a receiver record with observed values from "similar" donor records (Andridge and Little, 2010). The `microimpute` implementation uses the unconstrained distance hot deck approach from the StatMatch R package (D'Orazio et al., 2021). To identify the best match, the method computes distances between each receiver observation and all donor observations based on the common variables X, selects donor records within a specified distance threshold, and randomly samples from the eligible donors, with optional weighting by survey weights. For continuous and mixed covariates, Mahalanobis or Gower distance metrics can capture similarity across different variable types. The donated value will thus be an actual observed value from the donor file, ensuring plausibility.

Hot deck methods are nonparametric and avoid distributional assumptions, making them robust when the true conditional distribution is unknown or complex (D'Orazio et al., 2006). However, they face limitations:

- **Donor scarcity**: For receiver observations in sparse regions of the covariate space suitable donors may be unavailable, which can lead to the reuse of the same donors, which can affect the accuracy in the distribution of imputed values
- **Discrete outputs**: Imputed values are restricted to the set of observed donor values, which may not adequately represent the continuous nature of Y
- **Tail behavior**: Extreme values in the receiver population may have no appropriate donors or poor matches, leading to biased imputations at the tails

2.4.2 Ordinary Least Squares (OLS)

OLS imputation models the conditional mean of Y given X through linear regression:

$$E[Y|X] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \tag{2}$$

The coefficients $\boldsymbol{\beta}$ are estimated from the donor data by minimizing squared residuals. For stochastic imputation, the `microimpute` implementation, which builds on the `statsmodels` library, assumes normally distributed errors:

$$Y|X \sim \mathcal{N}(\boldsymbol{X}'\hat{\boldsymbol{\beta}}, \hat{\sigma}^2) \tag{3}$$

where $\hat{\sigma}^2$ is the estimated residual variance. Quantiles of $Y|X$ can then be computed as:

$$\hat{Q}_\tau(Y|X) = \boldsymbol{X}'\hat{\boldsymbol{\beta}} + \hat{\sigma} \cdot \Phi^{-1}(\tau) \tag{4}$$

where $\Phi^{-1}(\tau)$ is the τ-th quantile of the standard normal distribution.

OLS is computationally efficient and well-understood, but its assumptions are often violated by economic variables:

- **Linearity**: Relationships between Y and X may be nonlinear
- **Homoscedasticity**: Variance often increases with the level of Y, particularly for wealth
- **Normality**: Heavy-tailed distributions violate the normal error assumption, leading to inaccurate quantile estimates (Von Hippel, 2007)

2.4.3 Quantile Regression

Quantile regression directly models conditional quantiles rather than the conditional mean (Koenker and Bassett, 1978). For quantile $\tau \in (0, 1)$, the model is:

$$Q_\tau(Y|X) = \beta_0(\tau) + \beta_1(\tau)x_1 + \cdots + \beta_p(\tau)x_p \tag{5}$$

The coefficients $\boldsymbol{\beta}(\tau)$ are estimated by minimizing the asymmetric quantile loss function:

$$\mathcal{L}_\tau(\boldsymbol{\beta}) = \sum_{i=1}^{n} \rho_\tau(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) \tag{6}$$

where the check function $\rho_\tau(u) = u(\tau - \mathbf{1}_{u<0})$ penalizes positive and negative residuals asymmetrically.

The `microimpute` implementation of Quantile regression also uses the `statsmodels` library. By fitting models at multiple quantiles, it can approximate the entire conditional distribution of $Y|X$. For imputation, a random quantile $\tau^* \sim \text{Uniform}(0, 1)$ can be drawn, and the imputed value computed by interpolating between estimated quantile functions (Wei et al., 2014).

Quantile regression offers advantages over OLS:

- **Robustness**: Less sensitive to outliers than least squares
- **Heteroscedasticity**: Naturally accommodates varying spread across the distribution
- **No normality assumption**: Does not require specifying an error distribution

However, quantile regression still assumes linear relationships at each quantile, limiting its flexibility for complex data structures (Meinshausen, 2006).

2.4.4 Mixture Density Networks (MDN)

Mixture Density Networks combine neural networks with mixture models to estimate flexible conditional densities (Bishop, 1994). The approach models $P(Y|X)$ as a mixture of Gaussian components:

$$p(y|x) = \sum_{k=1}^{K} \pi_k(x) \cdot \mathcal{N}(y \mid \mu_k(x), \sigma_k^2(x)) \tag{7}$$

where K is the number of mixture components, and the mixing coefficients $\pi_k(x)$, means $\mu_k(x)$, and variances $\sigma_k^2(x)$ are all functions of the input x, parameterized by a neural network.

The neural network outputs $3K$ values for each input: K unnormalized log-probabilities (transformed via softmax to obtain π_k), K means, and K log-variances (exponentiated to ensure positivity). The network is trained by maximizing the log-likelihood:

$$\mathcal{L} = \sum_{i=1}^{n} \log p(y_i|x_i) \tag{8}$$

For imputation, one can either sample from the mixture distribution or compute quantiles numerically by inverting the cumulative distribution function.

The `microimpute` implementation uses PyTorch Tabular (Joseph, 2021), which provides optimized architectures for tabular data. MDNs offer several advantages:

- **Multimodal distributions**: The mixture formulation can capture multiple modes in $P(Y|X)$
- **Flexible nonlinearity**: Neural networks can approximate complex relationships between X and distributional parameters
- **Full density estimation**: Provides a complete probabilistic model, not just quantiles

Nonetheless, its limitations include:

- **Data requirements**: Neural networks typically require larger sample sizes for reliable estimation
- **Hyperparameter sensitivity**: Performance depends on architecture choices, learning rate, and number of components
- **Computational cost**: Training is more expensive than parametric methods

2.4.5 Quantile Random Forests (QRF)

Quantile Random Forests extend Random Forests to estimate conditional quantiles (Meinshausen, 2006). A standard Random Forest builds an ensemble of decision trees, each trained on a bootstrap sample, with predictions averaged across trees. QRF modifies this by retaining all training observations in each terminal node rather than just their mean.

For a new observation with covariates x, QRF estimates the conditional distribution function as:

$$\hat{F}(y|X = x) = \sum_{i=1}^{n} w_i(x) \cdot \mathbf{1}_{Y_i \leq y} \tag{9}$$

where the weight $w_i(x)$ reflects how often training observation i falls in the same terminal node as x across trees in the forest. Specifically:

$$w_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbf{1}_{x_i \in L_b(x)}}{|L_b(x)|} \tag{10}$$

where B is the number of trees, $L_b(x)$ is the terminal node containing x in tree b, and $|L_b(x)|$ is the number of training observations in that node.

The τ-th conditional quantile is then:

$$\hat{Q}_\tau(Y|X = x) = \inf\{y : \hat{F}(y|X = x) \geq \tau\} \tag{11}$$

The `microimpute` implementation uses the `quantile-forest` package, which offers several advantages for statistical matching:

1. **Distribution preservation**: By modeling the entire conditional distribution, QRF captures skewness, heavy tails, and other distributional features without parametric assumptions (Meinshausen, 2006).

2. **Nonlinear relationships**: The tree-based structure automatically captures complex nonlinearities and interactions without requiring explicit specification (Breiman, 2001).

3. **Single model for all quantiles**: Unlike standard quantile regression, which requires fitting separate models for each quantile, QRF estimates all quantiles from a single trained forest.

4. **Robustness**: The ensemble approach and quantile focus make QRF less sensitive to outliers than mean-based methods (Tang and Ishwaran, 2017).

Nevertheless, QRF also has limitations:

- **Extrapolation**: Trees cannot extrapolate beyond the range of training data, potentially underestimating extreme quantiles when donor and receiver distributions differ

- **Discrete approximation**: The estimated distribution is discrete, with support limited to observed Y values in the training data

2.5 Current Practice in Microsimulation

Statistical matching and data fusion are fundamental operations in microsimulation modeling, yet the methods employed in practice have remained relatively unchanged for decades. A review of major tax-benefit microsimulation models reveals a strong reliance on traditional imputation approaches, primarily hot deck matching and OLS-based regression.

Hot deck matching remains the dominant approach in European microsimulation. EUROMOD, the EU-wide tax-benefit model, employs a multi-stage imputation procedure combining predictive mean matching with distance-based hot deck methods to integrate consumption data from Household Budget Surveys into its EU-SILC input data (Sutherland and Figari, 2013). The appeal of hot deck methods lies in their simplicity and the guarantee that imputed values are observed values from the donor file, ensuring plausibility. However, as discussed above, these methods struggle with tail behavior and may not adequately capture the full conditional distribution of the target variable.

In U.S. tax policy microsimulation, regression-based approaches are more common. The Tax Policy Center employs a two-stage probit and OLS procedure for wealth imputation, first predicting the probability of holding each asset type, then predicting amounts conditional on positive holdings (Nunns, 2012). Similarly, the Institute on Taxation and Economic Policy (ITEP) model relies on statistical matching between tax return data and the American Community Survey, supplemented with regression-based imputations from the Survey of Consumer Finances (Institute on Taxation and Economic Policy, 2023). These regression approaches assume linear relationships and normally distributed errors; assumptions that are frequently violated by economic variables with heavy tails and heteroscedastic relationships.

More recent microsimulation efforts have begun incorporating administrative data through record linkage rather than statistical matching (Abowd, 2018), but this approach requires access to restricted data and raises privacy concerns. For researchers working with publicly available survey data, statistical matching remains essential, creating a need for methods that can better capture complex distributional features.

The limitations of traditional approaches can be particularly acute for heavily-tailed variables where relationships with predictors vary across the distribution. Machine learning methods such as Quantile Random Forests offer a promising alternative, as they can capture nonlinear relationships, model the entire conditional distribution, and handle heavy-tailed data without restrictive parametric assumptions. Despite these advantages, QRF and similar methods have seen limited adoption in mainstream microsimulation practice. The `microimpute` package aims to lower barriers to adopting these more flexible methods by providing a unified framework for comparing traditional and machine learning approaches, enabling researchers to empirically evaluate which method best suits their specific data characteristics.

2.6 Supplemental Security Income as a Validation Case

The limitations of traditional imputation approaches in microsimulation have concrete consequences on downstream policy analysis. The impact of wealth imputation quality for U.S. households can be illustrated through the Supplemental Security Income (SSI) program. SSI is a federal means-tested program for aged, blind, and disabled individuals with strict resource limits ($2,000 for individuals, $3,000 for couples) that have remained unchanged since 1989 (Social Security Administration, 2023). The resource test requires household wealth and asset data that the CPS does not collect, making SSI eligibility determination dependent on imputed or heuristic wealth values.

In 2024, SSI served approximately 7.4 million recipients with $59.6 billion in federal payments (Social Security Administration, 2024). Because different wealth imputation methods produce different wealth distributions, they directly affect which simulated households pass the resource test and therefore the resulting estimates of SSI recipient counts and expenditure. This makes SSI a particularly informative validation case for comparing imputation approaches, as described in Section 4.4.

3 Data

The selected application to demonstrate the `microimpute` methodology involves transferring net worth from the Survey of Consumer Finances (SCF) onto the Current Population Survey (CPS).

3.1 Survey of Consumer Finances

The Survey of Consumer Finances, sponsored by the Federal Reserve Board, is a triennial survey providing detailed information on U.S. households' assets, liabilities, income, and demographic characteristics. Its dual-frame sample design includes a standard national area-probability sample and a list sample deliberately oversampling wealthy households to better capture the skewed wealth distribution (Barceló, 2006). The SCF is a benchmark for wealth imputation research due to its detailed financial data and the known complexities arising from its design and the nature of wealth. Item nonresponse in public-use SCF datasets is addressed by the Federal Reserve through a multiple imputation approach that generates five complete datasets with different imputed values, using sequential regression-based procedures that incorporate range constraints, logical data structures, and empirical residuals to preserve the complex multivariate relationships inherent in wealth data (Kennickell, 1998).

Specifically, we use the 2022 summarized SCF as our donor dataset.

3.2 Current Population Survey

The Current Population Survey, conducted jointly by the U.S. Census Bureau and the Bureau of Labor Statistics, is a monthly survey of approximately 60,000 U.S. households that serves as the primary source of labor force statistics for the United States. The CPS uses a multistage probability-based sample designed to represent the civilian non-institutional population; on average, each sampled household represents approximately 2,500 households in the population. The Annual Social and Economic Supplement (ASEC) extends the core survey with detailed annual income data, including earnings, unemployment compensation, Social Security, pension income, interest, dividends, and other income sources. However, despite its comprehensive coverage of income and employment, the CPS does not collect information on household wealth, assets, or liabilities, which are key variables needed for wealth-based policy analysis. This omission motivates the need for statistical matching to transfer wealth information from the SCF onto the CPS.

Specifically, we use the Enhanced CPS 2024 produced by PolicyEngine (PolicyEngine, 2024) as our receiver dataset. The Enhanced CPS extends the base CPS ASEC microdata through survey reweighting and calibration to administrative totals, improving its representativeness for tax-benefit microsimulation. The SCF 2022 financial variables are uprated to 2024 dollars using CPI adjustment factors to ensure consistency with the receiver dataset's reference year.

3.3 Comparative analysis and characteristics for imputation

The SCF and CPS differ fundamentally in sampling design: the SCF employs a dual-frame approach combining a standard area-probability sample with a list sample that deliberately oversamples wealthy households, while the CPS uses a multistage area-probability household design representative of the civilian non-institutional population. This difference means that the donor and receiver surveys weight different parts of the wealth distribution differently, making proper survey weight integration during model training essential for unbiased imputation. The two surveys share a core set of demographic and income variables—age, sex, race, number of children, employment income, interest and dividend income, and pension income—but differ in their detailed variable coverage: the SCF collects granular asset and liability data absent from the CPS, while the CPS captures labor force dynamics and program participation variables not available in the SCF. This partial overlap constrains the conditioning set available for imputation and underscores the importance of the conditional independence assumption discussed in Section 2.2. Together, the contrasting designs, the heavy-tailed nature of wealth, and the variable overlap constraints make the SCF-to-CPS imputation a particularly informative test case for comparing methods, while also being directly relevant to U.S. policy microsimulation, where researchers require comprehensive microdata combining income, demographics, and wealth to analyze the distributional effects of tax and benefit reforms.

4 Methodology

4.1 `microimpute` package implementation

`microimpute`[1] is an open-source Python framework for statistical matching that provides a unified interface for imputing variables across survey datasets using multiple methods. The package addresses a gap in the microsimulation and microdata research community: while several imputation approaches exist, researchers have lacked a standardized tool for systematically comparing their distributional performance on a given dataset and automatically selecting the best approach. In this study, `microimpute` serves both as a methodological contribution to the broader research community and as the analytical engine powering the wealth imputation experiments described below.

[1] Complete documentation, implementation details, and usage examples are available at https://policyengine.github.io/microimpute/.

4.1.1 Core capabilities

The package currently supports five imputation methods: Hot Deck Matching, Ordinary Least Squares Linear Regression (OLS), Quantile Regression, Quantile Random Forests (QRF), and Mixture Density Networks (MDN). This approach allows researchers to systematically evaluate which technique provides the most accurate results for their specific dataset and research objectives. Additionally, the package is designed to be modular, allowing for easy extension with additional imputation methods in the future.

4.1.2 Key features

The package provides several capabilities designed for the challenges of cross-survey imputation, each of which plays a specific role in the wealth imputation experiments presented in this paper:

1. **Survey data weights integration**: Handles survey data weights through sampling to ensure that models are trained on a donor data set representative of the true distribution.

2. **Method comparison and benchmarking**: Allows researchers to easily compare different approaches and automatically determine the method providing the most accurate results.

3. **Flexible methodological set-up**: Enables advanced usage through specified hyperparameter setting and tuning.

4. **Quantile-based evaluation**: Uses quantile loss calculations to assess imputation quality across different parts of the distribution.

5. **Autoimputation**: Provides an integrated imputation pipeline that tunes method hyperparameters to the specific datasets, compares methods, and selects the best-performing to conduct the requested imputation in a single function call.

4.2 Evaluation framework

Comparing imputation methods requires metrics that assess distributional accuracy, not just point prediction. We employ two complementary quantitative metrics, along with visual distributional assessments.

The primary evaluation metric is quantile loss (also known as pinball loss). For a target quantile $\tau \in (0, 1)$, quantile loss assigns an asymmetric penalty that depends on the sign of the forecast error:

$$\mathcal{L}_\tau = \rho_\tau(y - \hat{Q}_\tau(y|x)) = (y - \hat{Q}_\tau(y|x)) \cdot (\tau - \mathbf{1}_{y<\hat{Q}_\tau(y|x)}) \tag{12}$$

Under-prediction of an upper-tail quantile is penalized at rate τ, whereas over-prediction is penalized at rate $1 - \tau$. This asymmetric structure captures directional bias—a crucial feature when imputing highly skewed variables like wealth, where large under-estimates in the right tail must be discouraged more strongly than equal-sized over-estimates (Koenker and Bassett, 1978). Compared with mean-squared or mean-absolute error, which penalize errors symmetrically, quantile loss directly targets the conditional distribution and remains robust to outliers (Ghenis, 2018). Averaging over multiple quantiles $\tau \in \{0.1, 0.2, \ldots, 0.9\}$ provides a comprehensive measure of how well the method estimates the conditional distribution across its entire range. Lower average quantile loss indicates better distributional calibration. Because quantile loss is not scale-invariant, the absolute values reported in Section 5 reflect the scale transformation applied; we supplement raw values with relative improvement percentages to aid interpretability.

As a complementary metric, we use the Wasserstein distance (also known as the earth mover's distance) to measure overall distributional similarity between imputed and reference distributions. Formally, for one-dimensional distributions P and Q:

$$W_1(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du \tag{13}$$

where F_P^{-1} and F_Q^{-1} are the quantile functions. The Wasserstein distance captures the total "work" required to transform one distribution into another, making it sensitive to the magnitude of distributional differences. While quantile loss evaluates conditional predictions at specific quantiles, the Wasserstein distance provides a single summary measure of marginal distributional accuracy, which is particularly informative for heavy-tailed variables like wealth.

In addition to these quantitative metrics, we employ two visual assessment approaches. First, we compare the imputed wealth distributions against the weighted SCF donor distribution, which, given the SCF's household weights that account for survey representativeness, provides a reference for the true U.S. wealth distribution. Second, we evaluate the average imputed wealth by disposable income decile, assessing whether each method produces a plausible monotonically increasing wealth-income relationship consistent with economic expectations.

4.3 Experimental setup

The `autoimpute` function orchestrates the full imputation pipeline in a single call: it tunes hyperparameters for applicable methods using the Optuna framework (Akiba et al., 2019), evaluates all methods through k-fold cross-validation on the donor data, and automatically selects the best-performing one to impute onto the receiver dataset. Using this pipeline, we evaluate all five imputation methods—QRF, OLS, Quantile Regression, Hot Deck Matching, and MDN—for net worth imputation from the SCF to the CPS.

Prior to model training, we apply an inverse hyperbolic sine (asinh) transformation to the net worth target variable. The asinh function, $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$, approximates the natural logarithm for large positive values while remaining well-defined for zero and negative values. This is particularly important for wealth data, where a meaningful fraction of households report negative net worth (i.e., debts exceeding assets), making a standard logarithmic transformation infeasible. The transformation compresses the extreme right tail of the wealth distribution, stabilizing variance and improving model performance across all methods. After imputation, predictions are back-transformed via the inverse function $\sinh(\cdot)$ to recover values on the original dollar scale.

This experimental design implicitly assumes that wealth is missing at random (MAR) conditional on the shared predictors X—that is, the probability of observing wealth in the donor survey does not depend on wealth itself once we condition on the demographic and financial variables available in both datasets, as formalized in Section 2.2. This assumption is standard in statistical matching but may be violated in practice: wealthy households may be less likely to participate in surveys or may underreport assets, creating MNAR-like patterns that the shared predictors cannot fully capture. We return to this limitation and discuss partial sensitivity checks in Section 6.3.

To create a ground truth for evaluation, we employ cross-validation, splitting the SCF into 5 folds. For each method, `autoimpute` performs the following steps:

1. Mask the wealth values of one holdout fold at a time
2. Tune the hyperparameters of methods with flexible configurations to the SCF dataset, namely Matching[2], QRF[3], and MDN[4]
   (a) Hyperparameter tuning is conducted through the `optuna` package, which parallelizes hyperparameter searches to identify the combination minimizing the average quantile loss across all quantiles for the SCF dataset
3. Impute wealth (`networth`) onto the SCF holdout set using each method, with the remaining four folds as training data
   (a) SCF survey data weights (captured in the variable `household_weight`) are passed into `autoimpute` for sampling
   (b) Demographic and financial variables available in both surveys serve as predictors, after preprocessing for consistent variable names and encoding: `is_female`, `age`, `race`, `own_children_in_household`, `employment_income`, `interest_dividend_income`, `pension_income`
   (c) Each method performs imputation at 20 equally spaced quantiles
4. Compare the imputed values to the SCF holdout values, measuring quantile loss at each of the imputation quantiles
5. Average quantile loss across all folds and quantiles
6. Select the method with the lowest average quantile loss as the best performer
7. Retrain the selected method on the full SCF dataset and use it to perform the final imputation of net worth onto the CPS

[2]Hyperparameters: `dist_fun`, `k`.
[3]Hyperparameters: `n_estimators`, `min_samples_split`, `min_samples_leaf`, `max_features`, `bootstrap`.
[4]Hyperparameters: `num_gaussian`, `learning_rate`.

This end-to-end pipeline allows for direct comparison between methods on a common evaluation framework and provides a robust assessment of relative performance. The final imputation step uses the entire SCF as training data, maximizing the information available to the selected model.

The imputation from the SCF onto the CPS is then replicated for the remaining four methods with an identical setup: training on the full SCF, the same predictor variables, and imputations at the median quantile to avoid introducing bias toward a specific side of the distribution.

4.4 Downstream policy validation

While distributional metrics such as quantile loss and Wasserstein distance assess how well an imputation method recovers the statistical properties of the target variable, they do not directly capture whether differences between methods matter for downstream policy analysis. To address this, we introduce a policy validation layer that evaluates imputation quality through its effect on a concrete policy simulation.

We use the Supplemental Security Income (SSI) program as a validation case. SSI is a federal means-tested program that provides cash assistance to aged, blind, and disabled individuals with limited income and resources (Social Security Administration, 2023). Critically, SSI eligibility includes a resource test with limits of $2,000 for individuals and $3,000 for couples, which directly depends on household wealth data that the CPS does not collect.

In current practice, PolicyEngine's microsimulation model (PolicyEngine, 2024) handles the missing wealth data by defaulting `ssi_countable_resources` to zero for all CPS households, meaning every categorically eligible individual passes the resource test. This produces a substantial overestimate of SSI participation, as it assumes no household possesses sufficient resources to be disqualified. Without household-level wealth data, the model cannot capture how the actual distribution of resources among potential SSI recipients affects program outcomes.

Our validation experiment proceeds as follows: for each imputation method, we feed the model's imputed net worth values into PolicyEngine as `ssi_countable_resources`, replacing the default zero-resource assumption with household-level wealth data. We then simulate SSI eligibility and benefits under each imputation scenario and compare the resulting recipient counts and total expenditure against SSA administrative totals (7.4 million recipients and $59.6 billion in federal payments for 2024). This provides an applied policy validation that complements the statistical evaluation metrics described above.

A limitation of this approach is that SCF net worth includes assets excluded from SSI countable resources (such as the primary residence and one vehicle), so the imputed values serve as a proxy rather than a precise measure of SSI-relevant resources. Nevertheless, the comparison across methods and against administrative totals provides a useful relative assessment of how imputation choices affect downstream policy estimates.

4.5 Additional methodological analyses

Two supplementary analyses complement the main experimental framework described above. Appendix B presents a predictor importance analysis using leave-one-out and progressive inclusion experiments on the QRF model, identifying the relative contribution of each predictor variable to wealth imputation quality and informing the plausibility of the conditional independence assumption. Meanwhile, to understand how the superiority of certain models over others generalize to other contexts and datasets, appendix A reports a cross-dataset benchmarking exercise that evaluates `microimpute`'s five methods across six publicly available datasets from diverse domains, assessing the generalizability of method performance beyond the SCF-CPS use case.

5 Results

5.1 Imputation results

The cross-validation results demonstrate QRF's strong performance across the wealth distribution. With the asinh transformation applied to the net worth target variable, QRF achieved the lowest average quantile loss across all quantiles, outperforming OLS, Hot Deck Matching, Quantile Regression, and MDN. QRF's overall consistency across the entire distribution made it the best-performing method for wealth imputation in this setting.

`microimpute`'s automated hyperparameter tuning via the Optuna framework contributed to these results. The optimal QRF configuration identified through cross-validation included 202 trees, a minimum of 1 sample per leaf, a split threshold of 5 samples, approximately 95% of features considered at each split, and bootstrap sampling enabled. These parameters balanced model complexity with generalization capability, preventing overfitting while capturing the non-linear relationships between demographic predictors and wealth outcomes.

5.1.1 Quantile loss

Comparing all five models to each other, we can evaluate performance through quantile loss, as well as comparing the overlap of the imputed wealth distribution with the original SCF data.

We observe how QRF proves to be the best-performing model out of the five. It significantly outperforms Matching and OLS on average, while only being outperformed by QuantReg past the 80th quantile. The dashed lines in Figure 1 indicate each method's average quantile loss across all quantiles, clearly separating QRF and Matching from the remaining methods. Matching's performance presents the perfect opportunity to see the quantile loss's directional bias at work. Because Matching does not incorporate quantile information into its imputation process in any way, it will match donor and receiver units, and thus impute values identically regardless of the quantile at work. The laddering behavior observed above results from the fact that Matching's wealth imputations consistently underpredict relative to the true wealth values, and this property is increasingly favored for quantiles below the median, and increasingly penalized as quantiles approach the right end of the distribution.

Table 1 summarizes average quantile loss across all quantiles and cross-validation folds, with relative improvement computed against the worst-performing method (MDN) as a reference.

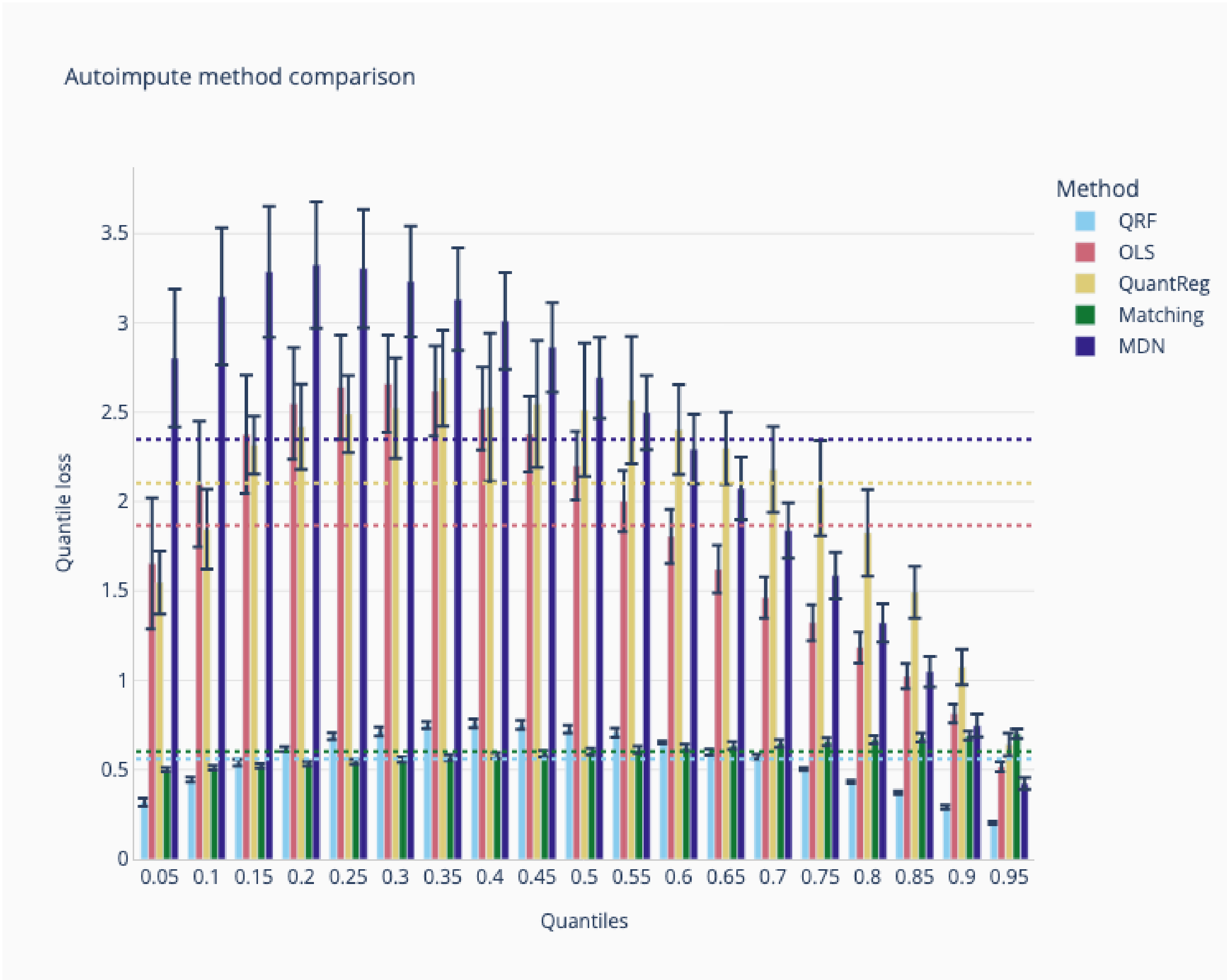

Figure 1: Cross-validation method performance at 20 equally spaced quantiles for all five imputation methods. Test quantile loss at each quantile is averaged across 5 cross-validation folds, and the dashed lines represent average quantile loss across all quantiles.

Table 1: Average quantile loss (asinh scale) across all quantiles and 5-fold cross-validation on the SCF test set. Relative improvement is computed as $(L_{\text{MDN}} - L_{\text{method}})/L_{\text{MDN}} \times 100\%$.

Method	Avg Quantile Loss	Std	Improvement (%)
QRF	0.560	0.016	76.2
Matching	0.601	0.018	74.4
OLS	1.866	0.196	20.6
QuantReg	2.104	0.241	10.4
MDN	2.349	0.227	—

QRF and Matching achieve substantially lower quantile loss than the remaining methods, with QRF reducing loss by 76.2% and Matching by 74.4% relative to MDN. The gap between these two leading methods is modest (0.560 vs. 0.601), while OLS, QuantReg, and MDN cluster at substantially higher loss values.

5.1.2 Imputed wealth distribution comparisons

By visually comparing the wealth distributions resulting from imputing with each method, and comparing them to the weighted donor distribution, we gain a more comprehensive understanding of imputation performance, moving past the test quantile loss average measured on the SCF. The distribution of wealth values imputed by QRF closely resembles the original SCF distribution, suggesting that QRF is not only the best method in terms of quantile loss but also strong when achieving distributional estimates close to the true wealth distribution in the United States. Because Matching directly takes values from the donor distribution and imputes them onto the receiver, it is unsurprising that its distribution closely resembles the donor distribution. However, Matching may struggle to uncover the non-linear relationship between the different predictors and wealth values, resulting in a seemingly correct marginal distribution but inaccurate imputation given certain household characteristics. Meanwhile, OLS, QuantReg, and MDN fail to capture the variability across the distribution and impute most values at the lower or higher ends of the distribution.

5.1.3 Wasserstein distance

Table 2 reports the Wasserstein distance between each model's imputed wealth distribution and the weighted SCF donor distribution, providing a single scalar summary of marginal distributional accuracy.

Table 2: Wasserstein distance (earth mover's distance) between each model's imputed net worth distribution on the CPS and the weighted SCF donor distribution. Lower values indicate closer distributional agreement.

Model	Wasserstein Distance
QRF	910,299
Matching	1,192,141
OLS	67,994,870
QuantReg	1.22×10^{14}
MDN	8.23×10^{14}

QRF achieves the lowest Wasserstein distance ($910,299), confirming that its imputed distribution most closely reproduces the donor wealth distribution. To contextualize this magnitude, the 2022 SCF reports a median U.S. household net worth of approximately $192,700 (Board of Governors of the Federal Reserve System, 2023); QRF's Wasserstein distance thus represents roughly 4.7 median household net worths of cumulative distributional displacement, which is small relative to the full range of the wealth distribution but non-trivial in absolute terms. Matching follows at $1.19 million, consistent with its mechanism

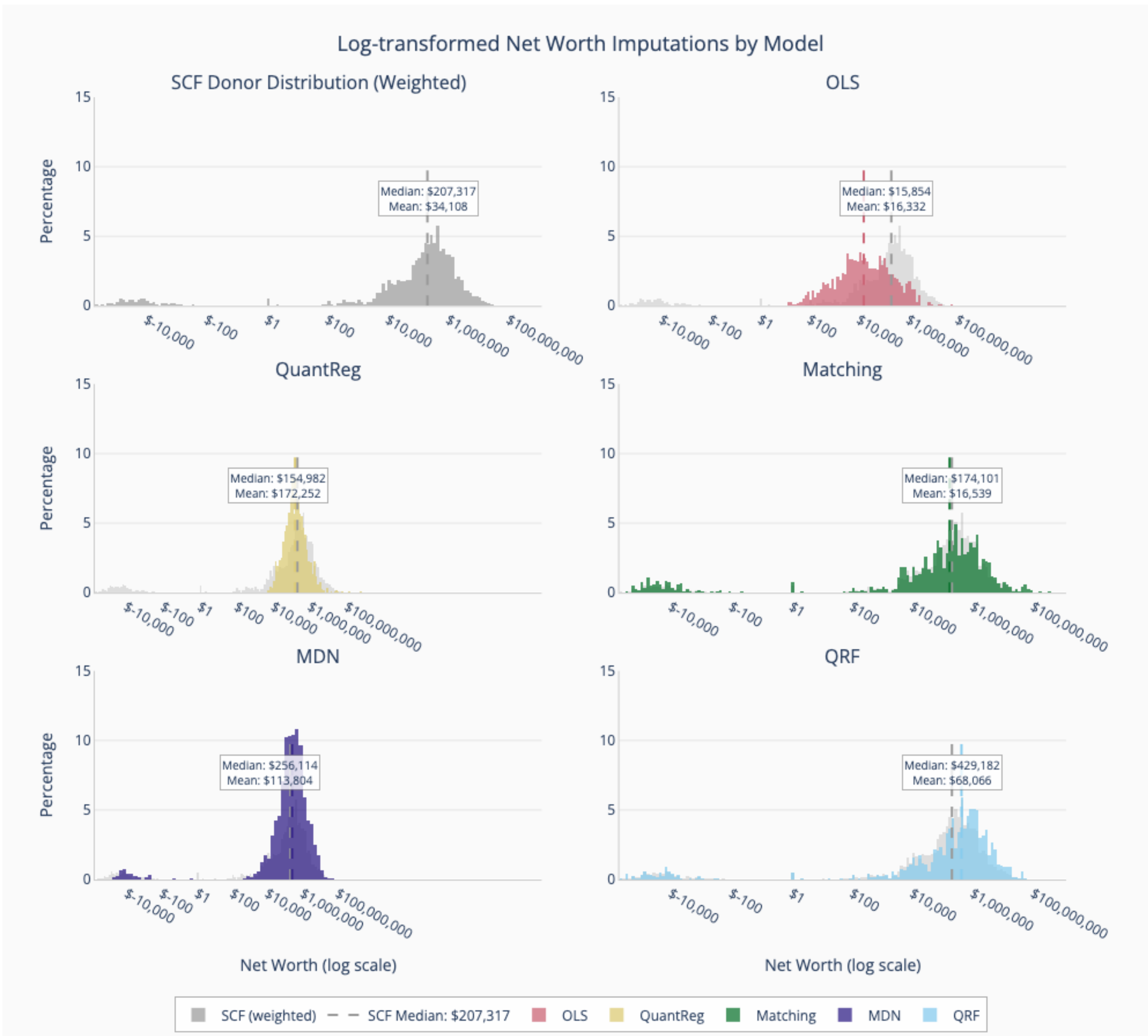

Figure 2: Log-transformed net worth distributions for all five imputation methods, compared to the weighted SCF donor distribution. Dashed lines represent median values.

of drawing values directly from the donor pool. OLS produces a substantially larger distance ($68.0 million), reflecting its compressed predictions that fail to span the full range of the wealth distribution. QuantReg and MDN exhibit Wasserstein distances several orders of magnitude larger, indicating severe distributional mismatch driven by concentrated imputed values in narrow regions.

5.1.4 Distribution of wealth by disposable income decile

Beyond the statistical metrics reported above, we perform a plausibility check by examining whether imputed wealth exhibits economically expected relationships with observable variables. Household wealth should increase monotonically with disposable income, reflecting the well-documented positive correlation between income and wealth accumulation. This check is particularly important for wealth imputation: a method that achieves competitive quantile loss but produces wealth-income patterns that violate basic economic relationships would be unreliable for downstream policy analysis, where the joint distribution of income and wealth drives eligibility and benefit calculations.

These results support the observations above, with QRF presenting the most consistent and plausible relationship to disposable income, with a monotonically increasing average wealth as deciles increase—a pattern consistent with economic expectations and serving as a plausibility check on the imputation. This plot also demonstrates the caveats of the other models, for example showing the lower variability in OLS's predictions, and Matching's consistent underprediction across deciles.

5.2 Policy validation: SSI simulation

Following the downstream validation framework described in Section 4.4, we feed each model's imputed net worth into PolicyEngine as `ssi_countable_resources` and simulate SSI eligibility and benefits. Table 3 compares simulated recipient counts and total expenditure against SSA administrative totals for 2024.

Table 3: SSI simulation outcomes by imputation method compared to SSA administrative totals (2024). The baseline scenario uses PolicyEngine's default behavior, where `ssi_countable_resources` defaults to zero for all CPS households.

Model	Recipients (M)	Expenditure ($B)	Recipients % of Admin
Admin total	7.4	59.6	100.0%
Baseline (no wealth)	19.8	152.6	267.2%
QuantReg	8.3	74.6	111.7%
MDN	10.4	85.4	140.0%
Matching	11.0	88.8	148.3%
OLS	11.3	100.5	153.3%
QRF	12.1	102.8	163.3%

Without imputed wealth data, the baseline scenario, where `ssi_countable_resources` defaults to zero for all CPS households, produces 19.8 million recipients and $152.6 billion

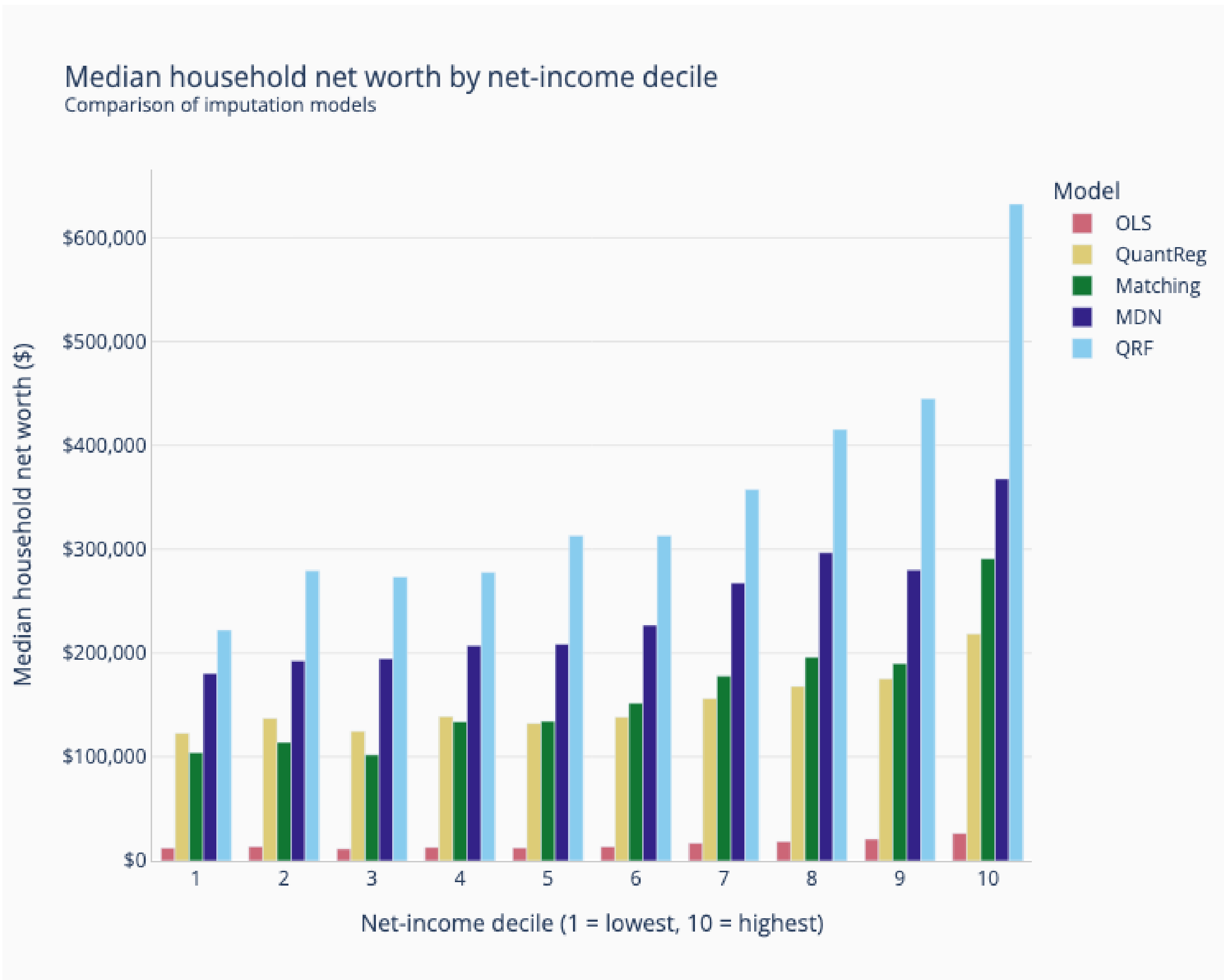

Figure 3: Average household net worth by disposable income decile model comparison. Households are divided into 10 equally sized groups based on their disposable income.

in expenditure (267.2% of administrative totals), substantially overestimating SSI program size. All five imputation methods substantially reduce this overestimation by screening out households whose imputed wealth exceeds the resource limits. Quantile Regression yields the closest estimates to administrative totals among the imputation methods (111.7% of admin recipients), followed by MDN (140.0%). QRF, despite achieving the lowest quantile loss in the cross-validation evaluation, produces higher SSI estimates (163.3%) than QuantReg or MDN in this policy validation, though still far closer to administrative totals than the baseline.

The divergence between statistical and policy performance merits further explanation. QRF, despite achieving the best quantile loss, produces higher SSI estimates than QuantReg (163.3% vs. 111.7%). This apparent paradox reflects the interaction between imputation accuracy and the policy function. QRF faithfully reproduces the conditional distribution of wealth, including the substantial mass of near-zero wealth values in the SCF; when mapped through the SSI resource test, these near-zero imputations place more households just below the eligibility threshold. QuantReg's linear compression of the wealth distribution, while statistically less accurate, may push more values above the threshold, coincidentally producing estimates closer to administrative totals. This underscores that proximity to administrative totals depends on both imputation quality and the specific policy function, and should not be treated as a standalone measure of imputation accuracy.

The remaining overestimation across all imputation methods also requires additional contex. The imputed variable is SCF net worth, which includes assets such as primary residence and vehicles that are excluded from SSI countable resources under program rules. Using total net worth as a proxy for countable resources therefore does not precisely replicate the SSI resource test, and will tend to disqualify fewer households than a measure limited to countable resources would.

6 Discussion

This paper has evaluated five imputation methods within the `microimpute` framework, finding that Quantile Random Forests offer meaningful advantages for the SCF-to-CPS wealth imputation application. By preserving the full conditional distribution of wealth, QRF maintains the key statistical properties of wealth data that traditional methods struggle to capture.

6.1 Strengths powered by `microimpute`

The `microimpute` package's design philosophy and implementation choices provide several key advantages that contributed to the success of our wealth imputation analysis, and will extend to future model comparisons across applications:

1. **Unified interface for method comparison**: `microimpute`'s consistent API across all imputation methods enabled systematic benchmarking without implementation-specific biases. This standardization ensures that performance differences reflect genuine methodological advantages rather than implementation artifacts (PolicyEngine, 2025).

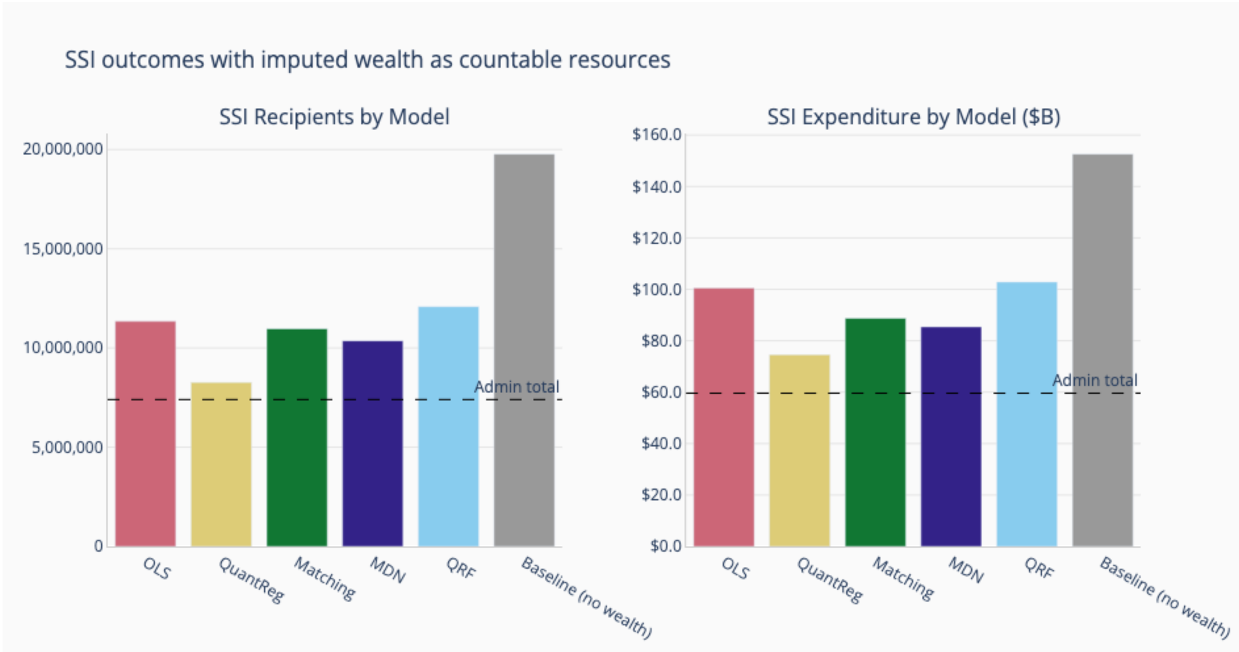

Figure 4: SSI recipients (left) and total expenditure (right) by imputation method, compared to the SSA administrative totals (dashed line) and the baseline scenario without imputed wealth.

2. **Automated method selection**: The package's `autoimpute` function streamlines the imputation workflow by automatically comparing methods and selecting the best performer based on quantile loss metrics. This feature proved particularly valuable given the wealth data's complexity, as it removed subjective method selection and ensured optimal performance.

3. **Survey weight integration**: `microimpute`'s native support for survey weights through stratified sampling ensures that imputation models properly represent population distributions. This capability is crucial when transferring information between surveys with different sampling designs, such as the SCF's oversampling of wealthy households.

4. **Quantile-aware evaluation**: By implementing quantile loss as the primary evaluation metric, `microimpute` directly addresses the challenges of skewed distributions. This metric's asymmetric penalty structure naturally prioritizes accurate imputation at distribution tails, where traditional metrics like Root Mean Squared Error (RMSE) often fail.

5. **Computational efficiency**: The package's optimized implementation enables processing of large microdata files while maintaining reasonable computation times. Cross-validation on the full SCF dataset, including QRF hyperparameter tuning, was completed in under 30 minutes on standard hardware, making iterative experimentation feasible.

6. **Open-source transparency**: As an open-source tool, `microimpute` allows full in-

spection and modification of imputation algorithms, promoting reproducibility and enabling custom extensions for specific research needs (PolicyEngine, 2025).

6.2 Generalizability and predictor sensitivity

The main SCF-CPS results are complemented by two supplementary analyses that inform the broader applicability of these findings.

The cross-dataset benchmarking exercise (Appendix A) reveals that method performance is context-dependent, with each method's strengths mapping to different data characteristics. QRF achieves the best Wasserstein distance on datasets exhibiting complex non-linear relationships, such as the Brazilian houses dataset and the SCF wealth imputation, where its ability to model flexible conditional distributions provides a clear advantage. Matching, by contrast, achieves the lowest Wasserstein distance on four of six benchmarking datasets and the best mean rank overall (1.67 vs. QRF's 1.83). This strong marginal distributional performance reflects Matching's core mechanism: by drawing imputed values directly from the donor pool, it inherently preserves the shape, skewness, and tail behavior of the original distribution, yielding low Wasserstein distances even when it may not optimally condition on predictor values. However, as the main SCF-CPS results demonstrate, Matching's distributional fidelity does not guarantee accurate conditional imputation. Households may receive plausible-looking wealth values that do not correctly reflect their individual characteristics. Thus, understanding the trade-offs between marginal and conditional accuracy, and evaluating how differences in survey design affect record matching becomes very important.

Meanwhile, OLS performs competitively on datasets where the predictor-target relationship is approximately linear (e.g., abalone, space_ga), highlighting that simpler parametric methods remain effective when their structural assumptions are met. MDN consistently ranks last across the benchmarking datasets, likely reflecting a combination of the relatively small sample sizes in these datasets and the hyperparameter sensitivity of neural network-based approaches, which require larger training sets to realize their flexibility advantage.

These findings reinforce the value of `microimpute`'s automated method comparison framework: no single method dominates across all settings, and the optimal choice depends on dataset size, the complexity of the predictor-target relationship, and whether the research objective prioritizes conditional accuracy or marginal distributional fidelity. Researchers should leverage the package's comparison tools to empirically determine the best approach for their specific data rather than defaulting to any single method.

The predictor importance analysis (Appendix B) reveals a clear hierarchy among the variables shared between the SCF and CPS. Financial predictors, particularly interest and dividend income, employment income, and age, account for the vast majority of imputation accuracy, with interest and dividend income alone more than doubling the quantile loss when removed. Demographic variables such as race and gender contribute relatively little to imputation quality. This concentration of predictive power in financial variables has two practical implications. First, the quality of wealth imputation rests primarily on the availability and accuracy of income-related variables in both surveys; researchers lacking financial predictors would see substantially degraded results. Second, diminishing returns from additional demographic predictors suggest that a parsimonious set of well-chosen financial variables can achieve most of the attainable accuracy, as confirmed by the progressive inclusion analysis showing that the first three predictors capture the majority of achievable performance. Nonetheless, the inclusion of demographic variables still provides incremental improvements, and their contribution to capturing nuanced wealth variation should not be overlooked.

`microimpute`'s support for predictor importance analysis and progressive inclusion experiments allows researchers to systematically assess which variables are driving imputation performance in their specific context, and to make informed decisions about variable selection when faced with data limitations.

Together, these supplementary analyses contextualize the main results: QRF's strong performance in the SCF-CPS setting reflects the particular characteristics of the wealth imputation problem—a large donor dataset, highly non-linear relationships, and informative financial predictors available in both surveys. Researchers applying `microimpute` to other settings should expect method rankings to vary with dataset characteristics and should leverage the package's comparison tools accordingly.

6.3 Limitations and future improvements

Despite the demonstrated advantages, there remain limitations worth considering for future development:

6.3.1 Current package limitations

While `microimpute` currently supports five imputation methods, expanding to include additional modern machine learning approaches such as more complex neural networks, gradient boosting machines, or deep learning architectures could further improve performance, particularly for complex multivariate relationships (Alaa et al., 2024). The package would benefit from implementing ensemble methods that combine multiple imputation approaches, potentially leveraging the strengths of different methods across different parts of the distribution. Moreover, model selection and assessment could be enhanced with evaluation metrics additional to quantile loss, ensuring a thorough understanding of each model's behavior at every step.

6.3.2 QRF-specific challenges

The terminal node sparsity issue identified in Section 2.4.5 remains a fundamental limitation of tree-based methods. When few training samples reach certain terminal nodes, multiple observations in the recipient dataset may receive identical imputed values, potentially underestimating variability in extreme wealth categories. Future work could explore adaptive tree construction methods that ensure minimum node occupancy or hybrid approaches that combine QRF with parametric methods at distribution extremes.

6.3.3 Missing data assumptions

All imputation methods in this study operate under the assumption that wealth is missing at random (MAR) conditional on the shared predictors X, as formalized in Section 2.2. This assumption may be violated in practice: wealthy individuals may be less likely to participate in surveys or may underreport assets, introducing missing-not-at-random (MNAR) patterns that the conditioning variables cannot fully account for (Andridge and Little, 2010). The predictor exclusion analysis in Appendix B provides a partial sensitivity check on the plausibility of the MAR assumption. The substantial degradation in imputation quality when financial predictors are removed, particularly the 116% increase in quantile loss when interest and dividend income is excludeds, demonstrates that the conditioning set is doing meaningful work in explaining wealth variation, which is consistent with MAR plausibility. However, this analysis cannot rule out residual dependence on unobserved factors. More formal MNAR sensitivity analyses, such as pattern-mixture models or selection models applied to the imputation framework, represent an important direction for future work. Researchers applying `microimpute` to settings where nonresponse or selection effects are suspected should consider supplementary diagnostics to assess the sensitivity of their results to departures from MAR.

These enhancements would position `microimpute` further as a comprehensive solution for survey data imputation challenges while maintaining its current strengths in ease of use and methodological rigor.

7 Conclusion

This paper has introduced `microimpute`, an open-source Python package for cross-survey imputation, and applied it to a substantive policy problem: imputing household wealth from the SCF onto the CPS. Through systematic comparison of five imputation methods using an inverse hyperbolic sine transformation to accommodate negative and zero net worth values, we find that Quantile Random Forests achieve the strongest conditional accuracy among the five methods evaluated for wealth imputation, driven by their capacity to model the complex non-linear relationships between demographic predictors and wealth. A downstream SSI simulation validates the practical importance of imputation method choice: without wealth data, the microsimulation overestimates SSI recipients by 167%, while imputation-based approaches substantially narrow this gap.

Our findings also reveal important nuances about method selection. Cross-dataset benchmarking shows that QRF's advantage reflects the specific characteristics of the wealth imputation problem rather than a universal superiority, and predictor importance analysis confirms that financial variables drive the majority of imputation accuracy.

These results carry two broader implications. First, imputation method selection should be empirically driven rather than prescribed; the optimal approach depends on dataset characteristics, and standardized comparison tools are essential for enabling this data-driven selection. Second, the gap between statistical and policy-relevant evaluation metrics—QRF achieves the best quantile loss yet not the closest SSI estimates to administrative totals—highlights the importance of validating imputation quality through downstream applications, not only through statistical benchmarks. Future work should expand `microimpute`'s method library to include gradient boosting and deep learning approaches (Alaa et al., 2024), explore ensemble strategies that combine methods across different parts of the distribution, and address the terminal node sparsity challenge that limits tree-based methods at extreme quantiles.

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A Cross-Dataset Benchmarking

To validate `microimpute` across diverse domains beyond the SCF-CPS wealth imputation use case, we conduct a cross-dataset benchmarking exercise using six publicly available regression datasets from the OpenML AutoML Benchmark regression suite (Vanschoren et al., 2014), spanning different domains, sample sizes, and dimensionalities.

A.1 Datasets

Table 4 summarizes the datasets used in this analysis.

Table 4: Datasets used for cross-dataset benchmarking.

Dataset	Domain	Rows	Target Variable
space_ga	Voting/demographics	3,107	ln(VOTES/POP)
elevators	Aircraft control	16,599	Goal
brazilian_houses	Real estate	10,692	total_(BRL)
onlinenewspopularity	News/media	39,644	shares
abalone	Marine biology	4,177	Class_number_of_rings
house_sales	Real estate	21,613	price

A.2 Methodology

For each dataset, we split the data 60/40 into donor and receiver subsets. The `autoimpute` pipeline runs internally on the 60% donor portion (cross-validation for method selection), and the selected method then imputes onto the 40% receiver subset. We evaluate imputation quality using the Wasserstein distance between the imputed and true distributions on the receiver subset. We use Wasserstein distance rather than quantile loss for this benchmarking because it provides a single summary measure of marginal distributional accuracy, which is more directly comparable across datasets with different scales and distributional shapes.

A.3 Results

Table 5 reports the Wasserstein distance for each method on each dataset, with lower values indicating better distributional recovery.

Table 6 summarizes the mean rank of each method across all six datasets.

Matching achieves the best Wasserstein distance on four of six datasets and the lowest mean rank overall (1.67), followed closely by QRF (1.83). QRF performs best on space_ga and brazilian_houses, the latter of which exhibits complex nonlinear relationships between predictors and house prices. OLS and Quantile Regression occupy middle ranks, while MDN consistently ranks last, likely reflecting a combination of the relatively small sample sizes in these datasets and the hyperparameter sensitivity of neural network-based approaches.

With only six datasets, the rank differences between methods—particularly the narrow gap between Matching (1.67) and QRF (1.83)—should be interpreted cautiously, as they

Table 5: Wasserstein distance by method and dataset. Bold values indicate the best-performing method for each dataset.

Dataset	Matching	QRF	OLS	QuantReg	MDN
space_ga	0.011	**0.010**	0.027	0.098	0.050
elevators	**0.0002**	0.0003	0.001	0.002	0.020
brazilian_houses	120.5	**27.2**	99.4	112.4	3,378.6
onlinenewspopularity	**239.9**	783.7	2,777.1	2,376.8	2,038.0
abalone	**0.003**	0.003	0.004	0.003	0.040
house_sales	**8,797.8**	12,576.6	41,862.3	64,725.3	544,730.4

Table 6: Mean rank across all six benchmarking datasets (lower is better).

Method	Mean Rank
Matching	1.67
QRF	1.83
OLS	3.33
QuantReg	3.67
MDN	4.50

are not statistically robust to the inclusion or exclusion of individual datasets. Notably, the relative rankings here differ from the main SCF-CPS analysis, where QRF achieves the lowest quantile loss. This is consistent with the expectation that method performance depends on dataset characteristics, and reinforces the value of `microimpute`'s method comparison framework, which allows researchers to empirically determine the best approach for their specific data rather than relying on a single method.

B Predictor Importance Analysis for SCF-CPS Imputation

To assess the relative contribution of each predictor variable to the quality of wealth imputation, we conduct two complementary analyses using the QRF model on the SCF data: a leave-one-out analysis and a progressive inclusion analysis. These analyses use six of the seven predictor variables from the main experimental setup; `own_children_in_household` is excluded because it was not available in the SCF preprocessing for this robustness analysis. The results help evaluate which shared variables between the SCF and CPS are most informative for imputing net worth.

B.1 Leave-one-out analysis

We measure the impact of each predictor by excluding it from the QRF model and observing the resulting increase in average quantile loss relative to the full model (baseline loss: 1,661,723, computed on the original dollar scale rather than the asinh-transformed scale used in Section 5). Table 7 reports the results.

Table 7: Leave-one-out predictor importance for QRF wealth imputation. Relative impact measures the percentage increase in average quantile loss when the predictor is excluded.

Predictor Removed	Avg Quantile Loss	Loss Increase	Relative Impact
interest_dividend_income	3,596,108	1,934,385	116.4%
employment_income	2,219,377	557,654	33.6%
age	2,189,845	528,122	31.8%
pension_income	1,875,120	213,397	12.8%
race	1,728,080	66,358	4.0%
is_female	1,698,452	36,729	2.2%
Baseline (all predictors)	1,661,723	—	—

Interest and dividend income is by far the most important predictor, with its removal more than doubling the quantile loss (116.4% increase). Employment income and age contribute substantially as well (33.6% and 31.8% increases, respectively). Pension income has a moderate effect (12.8%), while race and gender contribute relatively little to imputation accuracy (4.0% and 2.2%).

B.2 Progressive inclusion analysis

We complement the leave-one-out results by adding predictors one at a time, following the importance ordering determined by the leave-one-out analysis (Table 7), and measuring the marginal improvement at each step. Table 8 reports the results.

Table 8: Progressive predictor inclusion for QRF wealth imputation. Predictors are added in order of importance, and marginal improvement measures the reduction in average quantile loss from adding each predictor.

Step	Predictor Added	Avg Quantile Loss	Marginal Improvement
1	interest_dividend_income	5,037,628	—
2	age	2,752,098	2,285,529
3	employment_income	1,932,222	819,876
4	pension_income	1,737,197	195,025
5	race	1,704,094	33,103
6	is_female	1,685,442	18,652

The progressive analysis confirms the leave-one-out findings. The first three predictors (interest/dividend income, age, and employment income) account for the majority of imputation accuracy, with diminishing returns from additional demographic variables. All six predictors together yield the lowest quantile loss (1,685,442), indicating that each variable contributes positively, though the marginal gains from race and gender are small. The small difference between this value and the leave-one-out baseline (1,661,723) reflects variation across cross-validation fold assignments and random seeds between the two analyses.

B.3 Implications for the Conditional Independence Assumption

The strong predictive power of the shared financial variables, particularly interest and dividend income, employment income, and pension income, is consistent with, though not sufficient evidence for, the plausibility of the CIA in the SCF-CPS matching context. These variables capture key dimensions of household financial position that are available in both surveys. The fact that they collectively explain a large share of the variation in net worth suggests that the common variables X may be sufficient to render the target variable Y (wealth) approximately conditionally independent of variables Z unique to the CPS, though this assumption remains fundamentally untestable as discussed in Section 2.2.